Stratus Weather Server

Professional Weather Station Management Software

Stratus is a professional desktop application for Campbell Scientific weather station management, data collection, and real-time monitoring using the PakBus protocol.

Features & Capabilities

Campbell Scientific Integration

- **PakBus Protocol Support** - Native implementation of Campbell Scientific's PakBus protocol
- **Multi-Connection Types** - Serial (RS232/RS485), TCP/IP, GSM modems, LoRa, RF407
- **Data Collection** - Scheduled and on-demand data collection from datalogger tables
- **Program Management** - Upload, download, and manage CRBasic programs
- **File Operations** - Browse, transfer, and manage datalogger files
- **Clock Synchronization** - Automatic and manual clock sync with stations

Supported Dataloggers

- CR1000X, CR1000
- CR6
- CR3000
- CR800, CR850
- CR300
- CR200X

Communication Options

- **Serial** - RS232, RS485 direct connection
- **TCP/IP** - Ethernet, WiFi via NL121 or similar
- **RF** - RF407/RF412 spread spectrum radios

- **Cellular** - CELL200 series modems
- **LoRa** - Long-range IoT connectivity

Real-Time Monitoring Dashboard

Live Weather Display

- Real-time weather data visualization
- Connection health monitoring
- Data validation with automatic QC checks
- Alerts & alarms for data thresholds and connection issues

Solar Position Tracking

The dashboard displays real-time solar position calculated from station coordinates:

- **Sun Elevation** - Degrees above/below horizon (-90° to +90°)
- **Sun Azimuth** - Degrees from north (0° to 360°)
- **Nautical Dawn/Dusk** - When sun is 12° below horizon
- **Sunrise/Sunset** - Actual rise and set times
- **Day Length** - Hours of daylight

Calculations use NOAA solar position algorithms.

Air Density

Real-time air density calculated from:

- Temperature (°C)
- Pressure (hPa)
- Relative Humidity (%)

Uses the ideal gas law with humidity correction. Standard reference: 1.225 kg/m^3 at sea level.

Barometric Pressure

Dual display showing:

1. **Station Pressure** - Raw pressure at station altitude (hPa)
2. **Sea Level Pressure (QNH)** - Pressure calibrated to sea level (hPa)

Reference Evapotranspiration (ETo)

Calculated using FAO Penman-Monteith method (FAO-56 standard):

Required Inputs:

- Air temperature
- Relative humidity
- Wind speed
- Solar radiation
- Station altitude
- Station latitude

Outputs:

- Hourly ETo rate (mm/hr)
- Daily ETo (mm/day)
- Weekly/Monthly cumulative ETo

Battery Monitoring

Displays logger battery voltage with:

- Current voltage (V)
- Status indicator (Critical/Low/Fair/Good/Excellent)
- Charge percentage estimate
- 24-hour voltage history chart

Voltage Thresholds:

- Critical: < 11.5V
- Low: 11.5V - 12.0V
- Good: 12.0V - 13.5V
- Charging: > 13.5V

Wind Analysis (WMO/Beaufort Scale)

Wind Rose Charts

Traditional wind direction frequency distribution showing:

- **Wind Rose (Last 60 min)** - Recent wind patterns
- **Wind Rose (Today)** - Daily wind direction distribution
- **Wind Rose (Yesterday)** - Previous day comparison

Each rose displays direction frequency with speed-coded segments using WMO/Beaufort scale colors.

Wind Speed Scatter Plots

Individual wind speed observations plotted on a polar coordinate system:

- **Wind Speed (Last 30 min)** - Recent observations with tight clustering
- **Wind Speed (Today)** - All today's observations
- **Wind Speed (Yesterday)** - Previous day observations

Features:

- Points plotted by direction (angle) and speed (radius)
- Color-coded by wind speed according to WMO/Beaufort scale:
  - Light blue: Calm/Light (0-6 km/h)
  - Sky blue: Light Breeze (6-12 km/h)
  - Blue: Gentle Breeze (12-20 km/h)
  - Deep blue: Moderate (20-29 km/h)
  - Dark blue: Fresh (29-39 km/h)
  - Green: Strong (39-50 km/h)
  - Yellow: Near Gale (50-62 km/h)
  - Orange: Gale (62-75 km/h)
  - Red: Strong Gale (75-89 km/h)
  - Dark red: Storm+ (>89 km/h)
- Statistics: Average, Max, Min speed, Dominant direction
- Interactive tooltips showing exact values

Wind Compass

Real-time wind direction and speed display with compass rose.

Wind Power Analysis

Wind energy potential calculations for assessment purposes.

Data Management

Local Database

- **SQLite Storage** - All data stored locally for offline access
- **Data Export** - Export to CSV, JSON, or connect to external databases
- **Historical Charts** - View and analyze historical weather data
- **Backup & Restore** - Backup station configurations and data

Dashboard Export

Export the complete dashboard as a multi-page PDF report:

1. Click the **Export** button in the dashboard header
2. Select **Save as PDF**
3. The system will:
    - Capture the full dashboard with white background
    - Split content across multiple A4 pages
    - Add station name, date, and page numbers
    - Generate a downloadable PDF file

Station Setup

Adding a Station

1. **Open Station Manager** - Click `Station → Add Station` or press `Ctrl+N`

2. **Enter Basic Information**
   - Station Name: Descriptive name (e.g., "Main Campus Weather")
   - PakBus Address: Match datalogger setting (typically 1-4094)
3. **Configure Connection**

   For Serial:

   - Connection Type: Serial
   - COM Port: Select from dropdown
   - Baud Rate: 115200 (default)

   For TCP/IP:

   - Connection Type: TCP/IP
   - Host: IP address or hostname
   - Port: 6785 (default PakBus port)
4. **Security Settings** (if required)
   - Security Code: Enter datalogger security code
   - Leave empty if no security
5. **Test Connection**
   - Click "Test Connection"
   - Verify "Hello" response from datalogger
   - Check table definitions are retrieved
6. **Save Station** - Click "Add Station"

Data Collection Setup

Automatic Collection:

1. Select Station from the list
2. Go to Data Collection tab
3. Configure Schedule:
   - Table: Select table name (e.g., "WeatherData")
   - Interval: Collection frequency (e.g., every 5 minutes)

   - Mode: Most Recent or Since Last Collection
4. Enable Collection - Toggle "Enable Automatic Collection"

Collection Modes:

- **Most Recent Records** - Collects latest N records (good for real-time monitoring)
- **Since Last Collection** - Collects all new records since last time (ensures no data gaps)
- **Specific Range** - Collect records between dates/record numbers

---

Remote Access (Cloudflare Tunnel)

24/7 Public Access Features

- Secure HTTPS access via Cloudflare Tunnel
- Custom domain support (e.g., api.yourdomain.com)
- Auto-restart Windows services for reliability

Setup

```
# Run as Administrator
cd scripts

# 1. Initial tunnel setup
.\setup-cloudflare-tunnel.ps1 -Domain "yourdomain.com" -InstallService

# 2. Configure 24/7 production mode
.\setup-production-24-7.ps1

# 3. Check status
.\setup-production-24-7.ps1 -CheckStatus
```

Endpoints

- **Dashboard**: `https://yourdomain.com`
- **API**: `https://api.yourdomain.com`
- **Data Ingestion**: `https://api.yourdomain.com/api/weather-data`

Settings & Configuration

User Preferences

- **Temperature Unit** - Celsius or Fahrenheit
- **Wind Speed Unit** - km/h, m/s, mph, knots
- **Pressure Unit** - hPa, mbar, inHg, mmHg
- **Precipitation Unit** - mm or inches
- **Theme** - Light or Dark mode

Notification Settings

- Email notifications for alerts
- Push notifications
- Configurable temperature and wind thresholds

Data Processing Rules

Scaling:

- Scale Factor and Offset configuration
- Sensor-specific calibration coefficients

Quality Control:

- Min/Max bounds checking
- Rate-of-change limits
- Stuck sensor detection

Troubleshooting

Connection Issues

Problem	Solution

Connection Timeout	Check physical connection, verify IP/port, ensure datalogger is powered
Invalid Security Code	Verify code matches datalogger, check security level setting
PakBus Address Not Found	Verify address matches, check for conflicts, ensure port is enabled
COM Port Not Found	Check USB cable, install/update drivers, verify in Device Manager
Communication Error	Verify baud rate, check cable wiring (TX/RX crossover)

Data Collection Issues

Problem	Solution
Table Not Found	Verify table name in program, refresh table definitions
No New Data	Check datalogger scan interval, verify data is being logged
Host Unreachable	Check network connectivity, verify IP address, check firewall
Connection Refused	Verify PakBus/TCP is enabled, check port number (default 6785)

Best Practices

Connection Reliability

- Use static IP addresses when possible
- Configure reasonable timeout values
- Enable automatic reconnection

Data Integrity

- Use "Since Last Collection" mode for critical data
- Configure gap filling

- Regular backup of database

Security

- Enable PakBus security on remote stations
- Use VPN for cellular connections
- Limit API access to trusted clients

Performance

- Collect data at appropriate intervals
- Don't poll faster than datalogger scan rate
- Use efficient table structures

About

Stratus Weather Server
Version 1.0.0

Developer: Lukas Esterhuizen
Contact: esterhuizen2k@proton.me
License: MIT

Campbell Scientific and PakBus are trademarks of Campbell Scientific, Inc.

For additional help, visit Campbell Scientific Support at campbellsci.com/support